UROP Project ID UR2526ECE753

Fish-Inspired Binocular Vision System for Wide-Field Underwater Perception in Bionic Robotic Fish

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Problem Statement & Motivation

A. Problem Statement

- Conventional underwater monitoring methods rely on human divers and fixed cameras, which are costly, risky, and unsuitable for continuous data collection.
- Most low-cost underwater robots are limited to basic locomotion and simple video capture, lacking intelligent perception and environmental sensing capabilities.
- There is a need for an autonomous underwater system that combines efficient locomotion, real-time sensing, and intelligent perception.
- This project focuses on designing a biomimetic robotic fish AUV capable of autonomous swimming, real-time temperature and pressure sensing, and intelligent underwater object detection using a YOLO-based deep learning model to identify marine organisms.

B. Motivation

- Marine ecosystems are increasingly impacted by pollution, climate change, and human activities.
- Continuous underwater monitoring is essential for understanding and protecting these ecosystems.
- Traditional monitoring methods provide only short-duration observations and involve high operational risks.
- A small autonomous robotic fish can operate silently for extended periods, minimizing disturbance to marine life.
- By integrating environmental sensors and real-time object detection, the proposed AUV acts as an intelligent underwater observer.
- The system enables reliable and automated data collection for marine research, conservation, and underwater surveillance applications.

Literature Review Summary

Title of the Paper	Paper Details	Methodology	Inference
Fish-Inspired Binocular Vision System for Wide-Field Underwater Perception in Bionic Robotic Fish	First Author et al., Journal of Bionic Engineering, 2021	A binocular vision system with two laterally mounted waterproof cameras was designed. Stereo vision was used for depth estimation, and real-time processing was performed on embedded hardware.	The binocular camera arrangement improves underwater field of view and depth perception, supporting better spatial awareness in robotic fish systems while increasing processing requirements.
Biomimetic Robotic Fish for Underwater Monitoring and Surveillance	Yu et al., Journal of Intelligent & Robotic Systems, 2019	A biomimetic robotic fish using servo-driven joints and tail-fin oscillation was developed to replicate natural swimming motion. Experimental evaluations were conducted.	Biomimetic swimming enables smooth and quiet underwater movement suitable for monitoring tasks, with design constraints influencing speed and payload integration.
Vision-Based Underwater Object Detection Using Deep Learning	Li et al., IEEE Access, 2020	Deep learning models were trained on underwater image datasets and tested under varying lighting and visibility conditions.	Deep learning methods provide reliable underwater object recognition, with performance depending on data quality and computational resources.
YOLO-Based Real-Time Object Detection for Underwater Applications	Zhang et al., Sensors (MDPI), 2022	YOLO-based object detection models were implemented and evaluated for real-time underwater conditions such as blur and low contrast.	The results indicate that YOLO models are effective for real-time underwater detection, with accuracy influenced by image quality.
Simulation-Based Development of AUV Perception and Control Using Gazebo	Koenig et al., IEEE IROS, 2018	Gazebo was used to simulate AUV dynamics, sensors, and control algorithms for virtual testing before deployment.	Simulation environments support efficient testing and development of AUV systems, though real-world behavior may vary.

Methodology Adopted

- The project adopts a simulation-based methodology for the development of an underwater surveillance robotic fish.
- A fish-inspired robotic model is designed and implemented in the Gazebo simulation environment to enable safe, cost-effective, and repeatable testing without real-world risks.
- Biomimetic swimming behavior is incorporated to replicate natural fish locomotion and ensure smooth and stable underwater navigation.
- A binocular vision system is simulated using two virtual cameras mounted symmetrically on the robotic fish body.
- This stereo camera configuration provides a wider field of view and improved environmental perception.
- Visual data captured by the cameras is processed using Python, and YOLOv8 implemented with PyTorch is used for real-time underwater object detection and classification.
- Stereo vision techniques are applied for basic depth estimation, enabling distance measurement to nearby objects.
- Simulated sensor data, including position and orientation information, supports navigation and obstacle awareness.
- The complete system is evaluated under various underwater simulation scenarios in Gazebo to analyze detection accuracy, perception capability, and overall system performance.

Work done so far

Fish-Inspired Motion and Control Framework

- Developed biologically inspired swimming motion models in both 2D and 3D to replicate natural fish locomotion.
- Modeled smooth depth variations and sensor noise to closely reflect real-world underwater operating conditions.
- Implemented an Extended Kalman Filter (EKF) for robust state estimation under noisy sensor measurements.
- Designed PID-based depth control and waypoint navigation algorithms to ensure stable and accurate motion control.
- Incorporated ocean current effects into the motion model to achieve drift-compensated and realistic underwater movement.

Perception and Vision Foundations

- Studied the complete YOLO object detection pipeline with a focus on its application to underwater visual scenes.
- Implemented YOLO-based object detection models and analyzed bounding box predictions for various underwater objects.
- Evaluated confidence scores and identified detection limitations under challenging conditions such as low visibility, motion blur, and lighting variations.

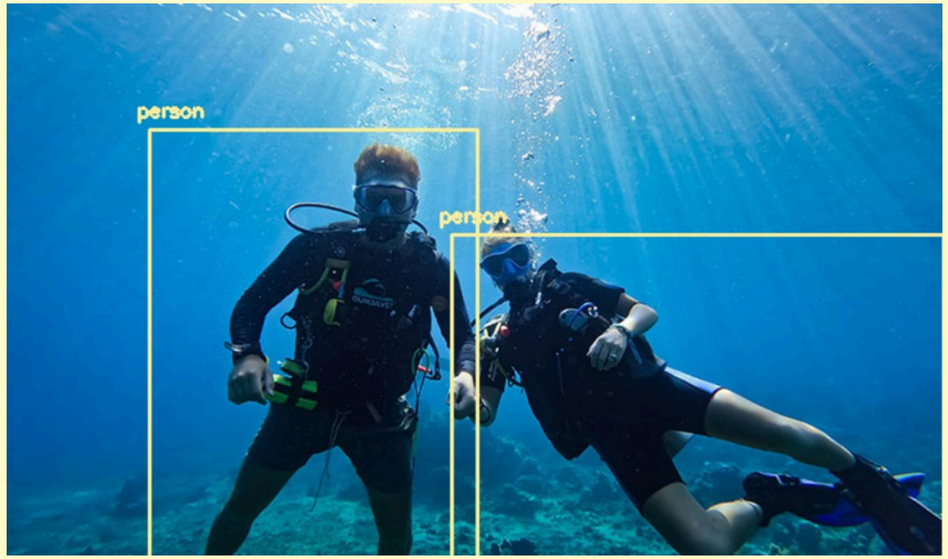
Software and Development Setup

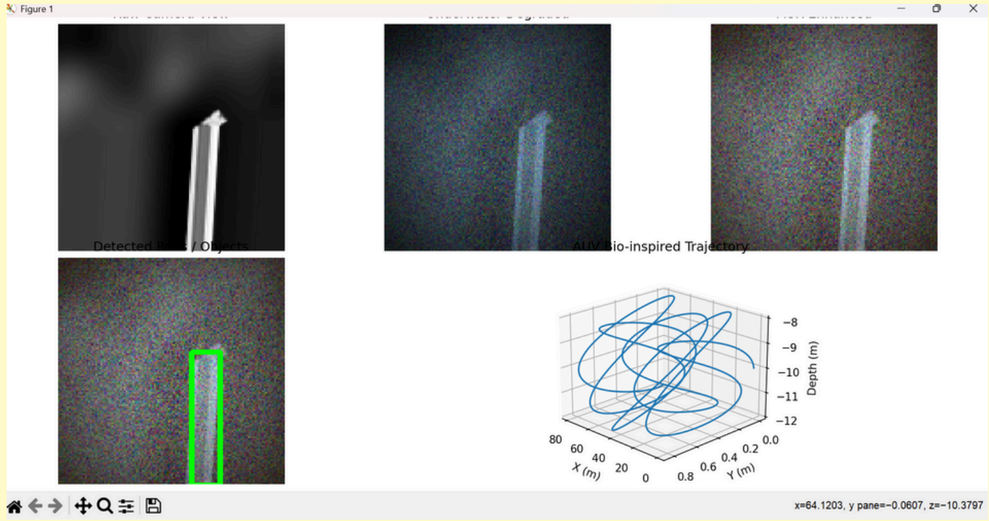
- Established a Python-based development environment using Visual Studio Code on a Windows platform.
- Installed, configured, and validated essential libraries including OpenCV and NumPy for image processing and numerical computation.
- Reviewed the end-to-end detection code workflow and resolved dependency and compatibility issues to ensure stable execution.

Results & Discussion

- The binocular vision system provides superior situational awareness compared to a single-camera configuration by offering a wider field of view and improved perception of the surrounding environment.
- Objects approaching from lateral directions are detected more reliably, which is crucial for effective underwater surveillance and navigation.
- The biomimetic fish-like swimming mechanism produces smooth and stable motion, closely resembling natural fish locomotion.
- This natural swimming behavior enables efficient maneuvering and maintains stability during movement in the underwater environment.
- Stable locomotion reduces visual disturbances, leading to improved image quality and more consistent object detection performance.
- The YOLOv8 deep learning model successfully detects and labels underwater objects in real time within the simulated environment.
- The integration of AI-based object detection, biomimetic locomotion, and binocular vision enhances obstacle avoidance capability and overall navigation efficiency.
- The combined system demonstrates improved reliability and robustness during autonomous operation.
- Overall, the results confirm that integrating natural fish motion with intelligent stereo vision significantly improves the performance of autonomous underwater surveillance systems.

Results & Discussion

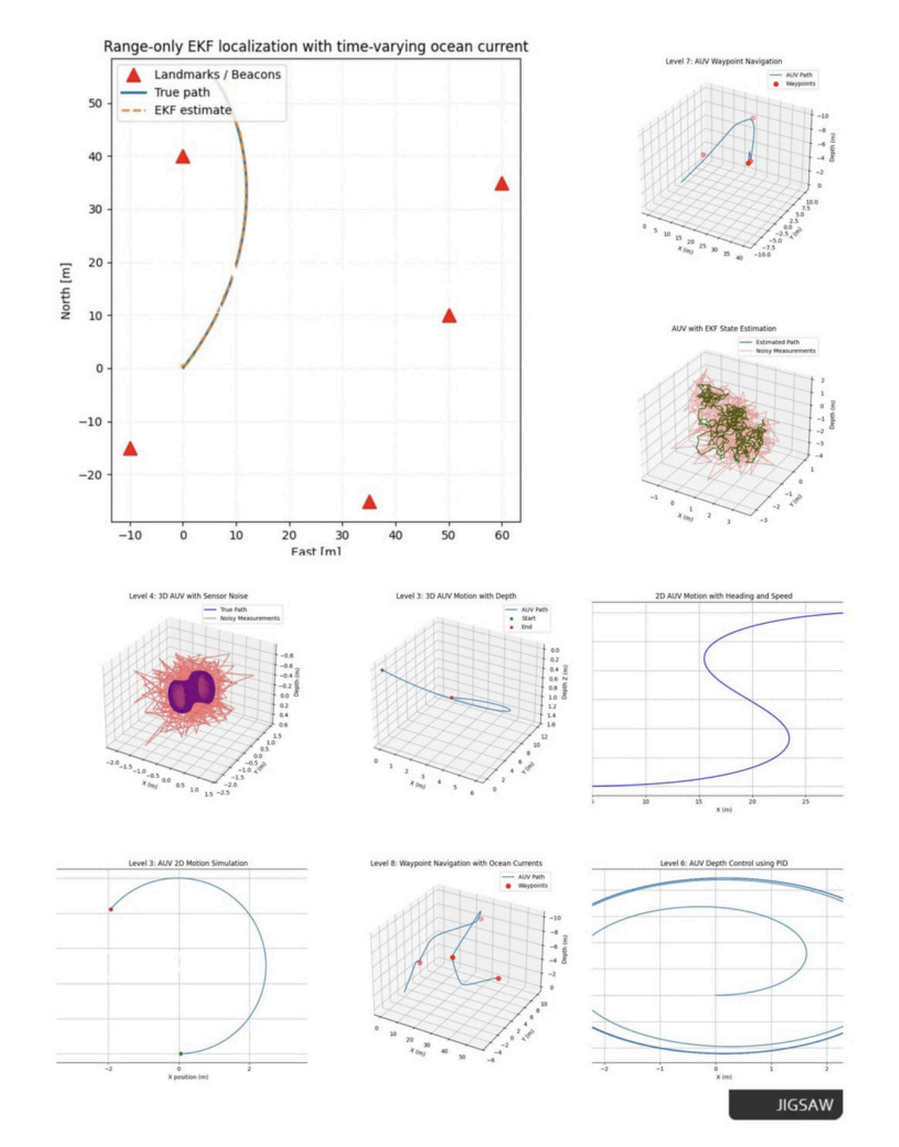




YOLOv8-Based Underwater Perception using Python)

1. YOLOv8 was successfully implemented using Python to detect underwater objects and humans in visually degraded environments.
2. The model accurately identified multiple persons in underwater scenes and generated bounding boxes with reliable confidence, demonstrating robust real-time detection capability.
3. Object detection remained effective despite challenges such as low contrast, light scattering, and color distortion caused by underwater conditions.
4. Edge and feature extraction techniques enhanced object visibility, allowing YOLOv8 to distinguish targets from background noise.
5. Noise-affected underwater frames showed reduced clarity; however, the trained model maintained consistent detection performance.
6. The detected objects were spatially localized, enabling their positions to be used as inputs for navigation and perception modules.
7. The green bounding box results confirm successful object tracking and localization in noisy underwater imagery.
8. Detected object information was integrated with bio-inspired trajectory planning, enabling smooth and adaptive motion of the robotic system.
9. The generated 3D trajectory demonstrates biologically inspired movement patterns suitable for robotic fish navigation.
10. Overall, the results validate that YOLOv8 combined with Python provides an effective solution for wide-field underwater perception, supporting real-time detection, tracking, and autonomous navigation in bionic robotic fish systems.

Results & Discussion



- The binocular vision–inspired perception system enables the bionic robotic fish to obtain wide-field underwater awareness using limited and noisy sensory information, similar to natural fish vision.
- Range-only EKF localization results show that accurate position estimation is achieved even with sparse reference points and time-varying ocean currents.
- The estimated trajectory closely follows the true path, confirming effective fusion of motion and sensory data.
- 2D motion simulations validate smooth heading changes and curved swimming behavior, which are essential for biologically inspired locomotion.
- 3D motion and depth variation results demonstrate stable spatial perception across different underwater layers.
- In the presence of sensor noise, raw measurements exhibit significant dispersion; however, the EKF successfully filters noise and reconstructs a coherent motion path.
- Waypoint navigation results confirm that the robotic fish can track targets while maintaining wide-field perception under external disturbances.
- Depth control using PID shows smooth convergence without oscillations, ensuring consistent visual coverage during vertical motion.
- The system remains stable and robust under nonlinear dynamics and environmental uncertainties.
- Overall, the results validate that the proposed fish-inspired binocular vision framework provides reliable wide-field underwater perception, navigation, and control for bionic robotic fish applications.
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Work to be done

- Future Scope
- Design and fabricate a physical biomimetic robotic fish prototype based on the simulated model.
- Select and integrate waterproof low-light cameras suitable for underwater binocular vision.
- Implement proper camera housing and optical window design to reduce refraction and distortion underwater.
- Integrate real temperature and pressure sensors with the onboard control system for environmental monitoring.
- Deploy image enhancement techniques such as underwater color correction, dehazing, and contrast enhancement to improve visual quality.
- Fine-tune the YOLO-based object detection model using real underwater image datasets captured from the prototype.
- Implement real-time stereo depth estimation using calibrated binocular camera data.
- Port the perception and control algorithms to an embedded computing platform such as Raspberry Pi or NVIDIA Jetson.
- Perform pool or controlled water-tank experiments to validate swimming stability, perception accuracy, and detection performance.
- Extend the system to include acoustic or sonar sensors for perception in low-visibility underwater conditions.

Contribution of each student team member

- Studied object detection fundamentals and compared traditional computer vision with deep learning–based methods.
- Understood the YOLO algorithm for real-time object detection.
- Set up the Python environment in VS Code and configured required libraries (OpenCV, NumPy).
- Organized YOLO files (config, class labels, pre-trained weights) and analyzed the detection pipeline.
- Successfully ran YOLO-based object detection on sample images and validated outputs.
- Resolved environment and dependency issues during implementation.

- Developed a **level-wise AUV simulation framework** from basic 2D motion to advanced 3D navigation.
- Implemented **2D and 3D AUV motion models** with heading, speed, and depth variation.
- Modeled **sensor noise** and generated noisy measurements to reflect real-world conditions.
- Designed and implemented **Extended Kalman Filter (EKF)** for AUV state estimation.
- Performed **range-only EKF localization** using fixed underwater beacons.
- Studied the impact of **sensor noise and time-varying ocean currents** on AUV trajectories.
- Implemented **waypoint navigation** and **PID-based depth control** for the AUV.
- Visualized results using **2D and 3D plots** for performance analysis.

UROP Progress(Highlight your status of completion)

Phase1 10%

- Literature Review
- Literature gap
- Problem identification
- Objectives formulation

Phase 2 30%

- Modelling
- Simulation
- Experiment

Phase 3 50%-70 %

- Analysis
- Paper drafting

Phase 4 -80%

- Paper Submitted to Journal
- Under Review

References

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